Do Metabolic Signaling Dynamics Predict Macrophage Polarization Fate?

UC DAVIS MOLECULAR AND CELLULAR BIOLOGY

Marion Hardy, Daniel Oberbauer, Michael Pargett and John G. Albeck · Department of Molecular and Cellular Biology, University of California, Davis

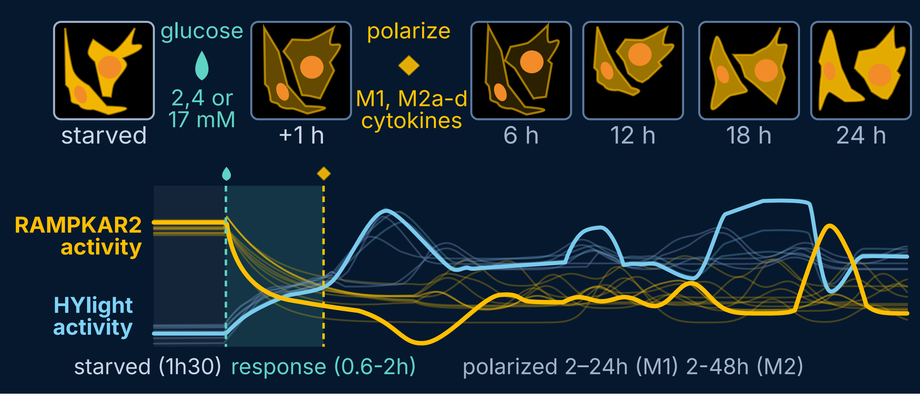
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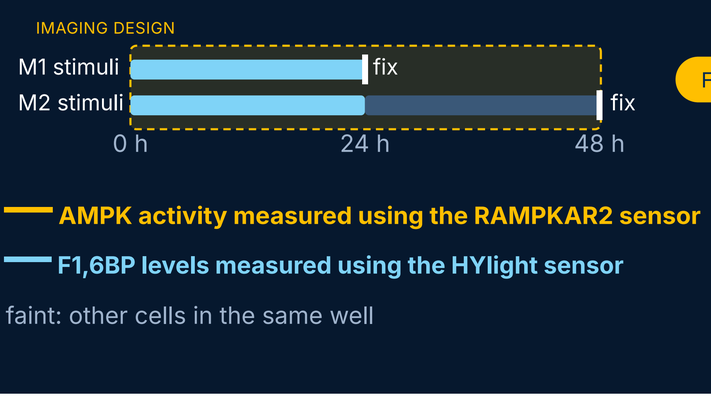
Chimeric antigen receptor macrophages (CAR-M) are being developed against solid tumors, where macrophages can infiltrate tissue that T cells struggle to reach. Their anti-tumor activity depends on holding an inflammatory state inside a tumor microenvironment that biases the other way [1]. That environment is nutrient-poor, especially low in glucose, and contains immunosuppressive signals such as IL-10, TGF-β and glucocorticoids. Approaching this problem through immunometabolism offers a possible lever: in the classical view, inflammatory activation shifts macrophages toward glycolysis, while anti-inflammatory behaviors rely on oxidative phosphorylation [2]. If that is true, then a cell's metabolic state could be used to predict or bias its functional state.

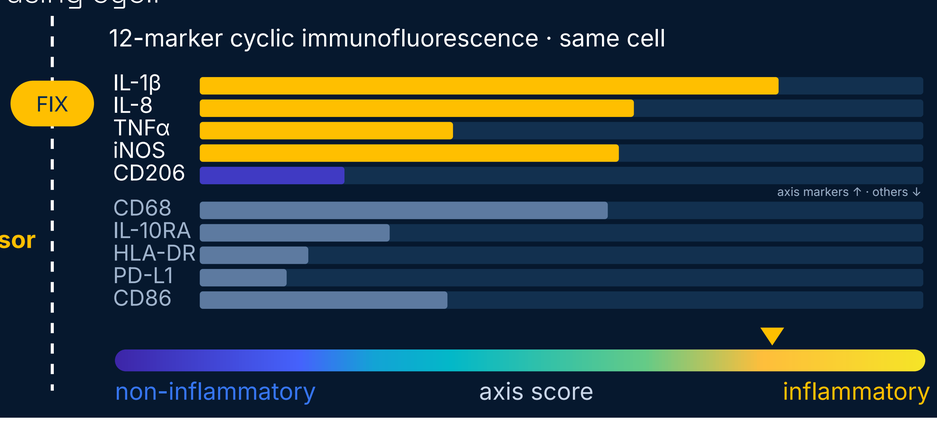
This picture, however, is built mostly from population averages measured at a single time point. Live-cell imaging of signaling dynamics has shown that genetically identical cells exposed to the same stimulus respond differently, and that the timing and shape of a response can carry information a snapshot misses [3-4]. Using genetically encoded biosensors, the Albeck lab has shown that glycolytic flux and AMPK activity are also heterogeneous and dynamic from cell to cell. Whether this metabolic variability matters for what an individual cell becomes is unknown.

To test this, we imaged THP-1-derived macrophages expressing HYlight (fructose-1,6-bisphosphate, a readout of glycolytic activity) and RAMPKAR2 (AMPK activity) [5-6] through glucose starvation, re-feeding and polarization. The cells received five stimuli, from inflammatory (IFNγ + LPS) to suppressive (IL-10 + TGF-β + hydrocortisone), at glucose concentrations from 2, 4 or 17 mM. We then stained the same cells for 12 markers by cyclic immunofluorescence [7]. We summarized each cell's endpoint as a single inflammatory score and asked two questions. First, between treatments, do the more glycolytic conditions end up more inflammatory? Second, cell by cell, do live metabolic dynamics predict where a macrophage lands, beyond what its treatment already sets?

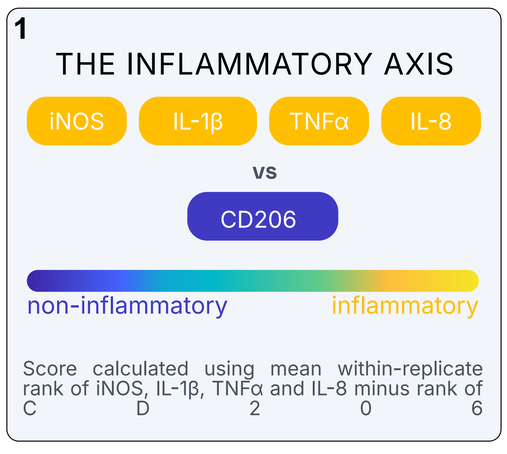
METHODS Glucose starve + feed → polarize → image AMPK and F1,6BP for 24/48 h → fix and image markers using cycIF







THE INFLAMMATORY AXIS explains a fifth of the variance across stimuli but varies most within wells.



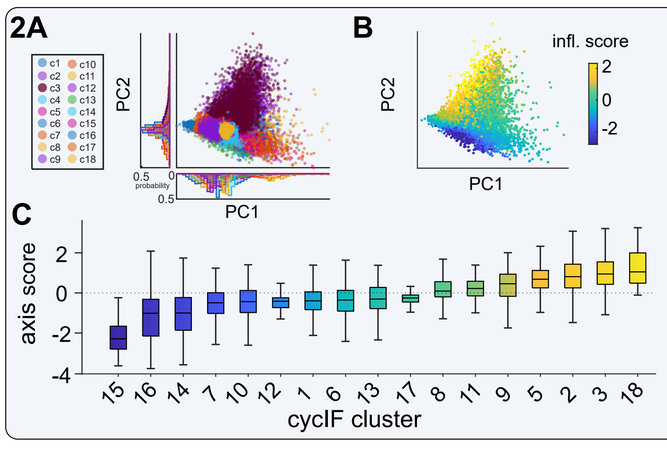


Figure 2. Individual cellular endpoint profiles cluster along the inflammatory axis. (A) Single-cell cycIF profiles are plotted on a PCA using all 10 functional markers, clustered. (B) Cells were scored along the inflammatory axis. (C) Clusters median scores (≥ 50 cells), are calculated and ordered from lesser to highest score. η^2 = 46.4% pooled, 37.8% within condition and glucose effect removed. Data was analyzed for 27 521 cells, with full live cell traces and >90% mapped stain markers (representing 67% of all acquired data). In total, 3 experimental replicates were run, each with 2 technical replicates.

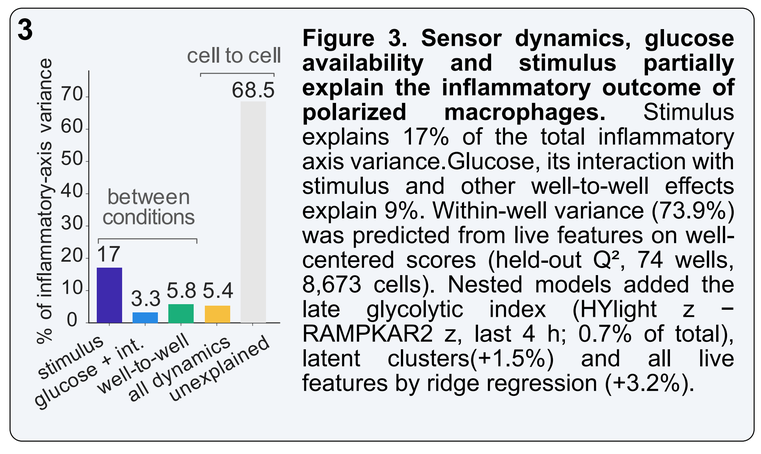
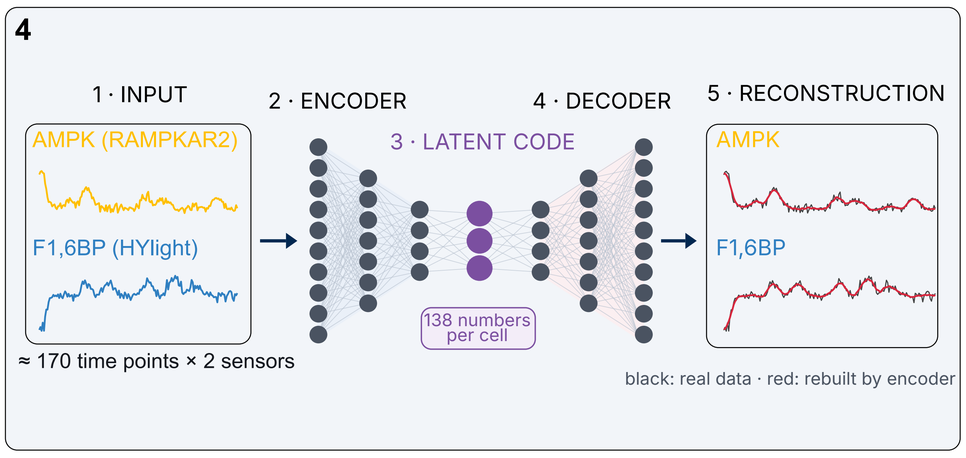
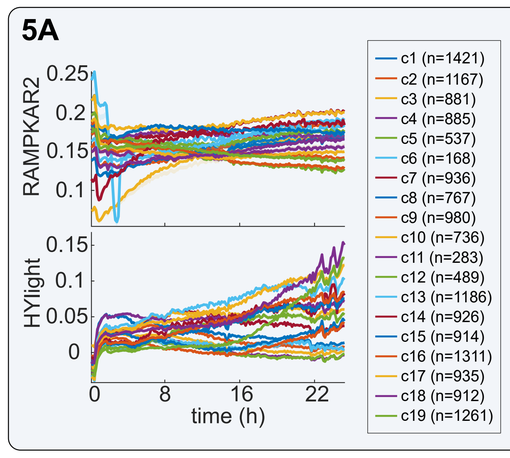


Figure 3. Sensor dynamics, glucose availability and stimulus partially explain the inflammatory outcome of polarized macrophages. Stimulus explains 17% of the total inflammatory axis variance.Glucose, its interaction with stimulus and other well-to-well effects explain 9%. Within-well variance (73.9%) was predicted from live features on well-centered scores (held-out Q^2, 74 wells, 8,673 cells). Nested models added the late glycolytic index (HYlight z − RAMPKAR2 z, last 4 h; 0.7% of total), latent clusters(+1.5%) and all live features by ridge regression (+3.2%).

READING LIVE DYNAMICS using a latent auto-encoder algorithm to classify cell traces into different groups.





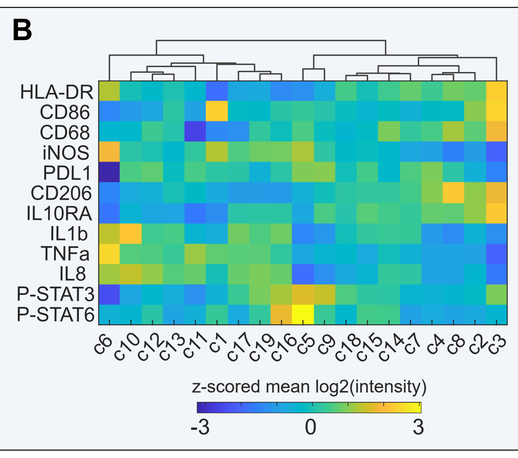
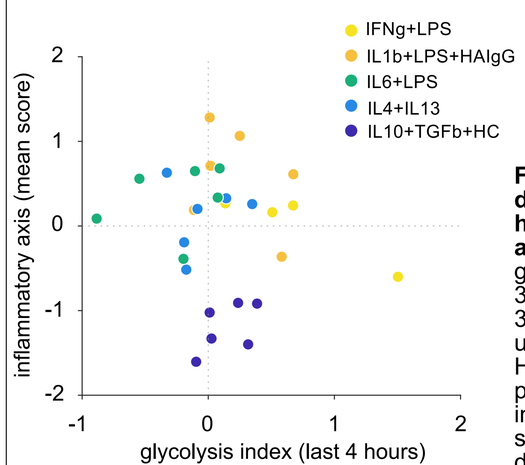


Figure 5. Latent space encoded dynamic clusters have their own endpoint stain fingerprint. (A) A time series clustering auto-encoder compressed each cell's pair of traces into a latent representation where Leiden clustering defined 19 clusters (c1–c19, n = cells per cluster). Mean RAMPKAR2 (AMPK activity) and HYlight (fructose 1,6 biphosphate levels) traces per cluster. (B) Heatmap of the mean log2 endpoint intensity of each cycIF marker, z-scored across clusters within each marker. Clusters are ordered by hierarchical clustering of these profile.

PREDICTING CYTOKINE BEHAVIOR by using metabolic parameters from the live cell dynamics.

AT THE POPULATION LEVEL



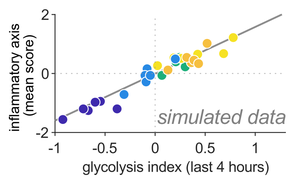


Figure 6. Glycolysis and inflammation are decoupled between conditions. IFNγ+LPS is highly glycolytic yet sits mid inflammatory axis. Each dot is one condition (stimulus x glucose) in one replicate, colored by stimulus (n = 30, 5 stimuli x 3 glucose levels x replicates 2 and 3; replicate 1 excluded because CD206 was undetected). The x-axis is the glycolytic index, HYlight z − RAMPKAR2 z: each sensor is z-scored per replicate and averaged over the last 4 h of imaging. The y-axis is the mean inflammatory-axis score of the same cells. Decoupled ordering is determined by Spearman ρ = −0.06, p = 0.76. HC, hydrocortisone; HAIgG, heat-aggregated IgG.

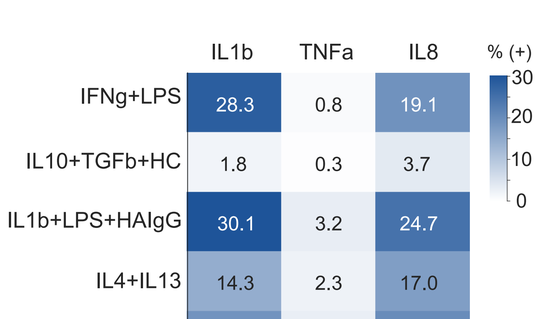
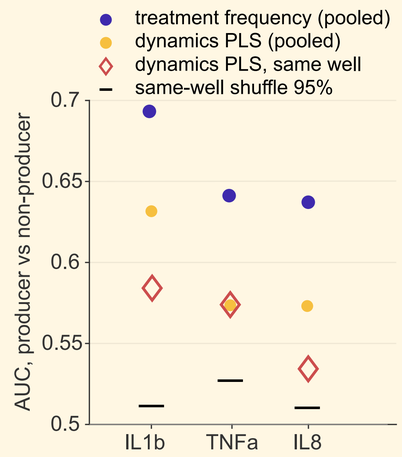


Figure 7. Stimuli set the frequency of cytokine producers. The heatmap shows the mean percentage of IL-1β⁺, TNFα⁺ and IL-8⁺ cells per well for each stimulus (rows) and cytokine (columns), pooled across glucose doses and replicates. Producers were called with a per-marker gate on background-corrected cycIF intensities. The gate is the median + 5 × robust SD of stained cells, set per replicate. Frequencies were computed per well: 120 wells, 3 replicates, 15 666 cells per cytokine (≈130 cells/well). The effect size η^2 = SS_stimulus/SS_total is the share of well-to-well variance in producer frequency explained by stimulus. Significance was tested by permuting stimulus labels across wells (* p = 0.002). Glucose has small effects: η^2 0.07 for TNFα (p 0.02), 0.06 for IL-8 (p 0.03), and none for IL-1β.

CELL BY CELL

Figure 8. Live dynamics carry a small amount of information about which cells in a well become cytokine producers. Each point is an AUC: the probability that a randomly chosen producer scores higher than a randomly chosen non-producer, where 0.5 is chance. Blue circles are a reference that scores each cell by the producer frequency of its treatment condition, estimated from the other wells, and so show how much treatment alone identifies producers. Yellow circles score cells by the out-of-fold prediction of a partial least squares (PLS) model built on the autoencoder latent of each cell's RAMPKAR2 and HYlight traces, pooled across all cells, so they still include between-well treatment information. Red diamonds also use PLS predictions but removes treatment, glucose and well effects. Black ticks mark the 95% null from shuffling scores within wells. Treatment is the strongest predictor of producer identity (AUC 0.69 for IL-1β, 0.64 for TNFα and 0.64 for IL-8). Within wells, dynamics still rank producers above their non-producer neighbours (AUC 0.58 for IL-1β, 0.57 for TNFα and 0.53 for IL-8), all above the shuffle null and IL-8 only marginally. The TNFα estimate is the least reliable, because its gate calls few cells (2.1% producers) and has a higher false-positive rate in secondary-only controls.



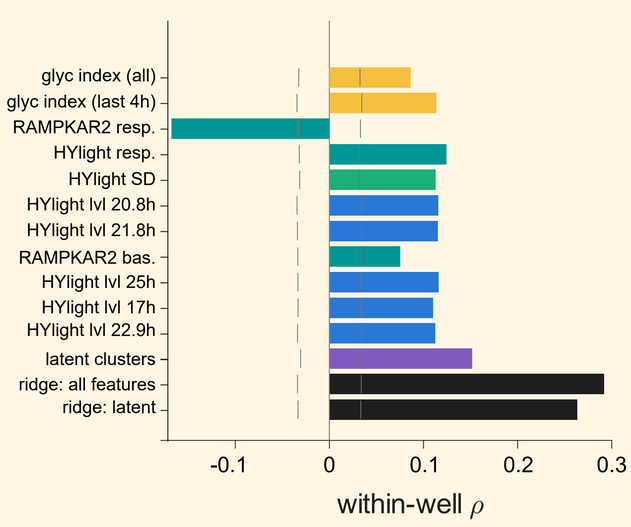


Figure 9. Live metabolic dynamics weakly but reproducibly track the inflammatory axis per cell. Cells with stronger AMPK response to starve tend to become more inflammatory. Each bar is the Spearman correlation (ρ) between one live-imaging feature and the inflammatory axis score. Features were ranked and models fitted on half of the wells. ρ is reported on the other half. Ticks mark the 95% null from shuffling cells within wells. The glycolytic index (yellow) averaged over the whole imaging window or its last 4 h. Glucose-response features (teal).The response is the mean signal over the first 1.5 h after glucose addition minus the starved baseline, and the baseline is the starved level itself. HYlight_sd (green) is the standard deviation of the trace. HYlight levels (blue) are mean HYlight in 1h bins over the last 8 h of imaging. Latent clusters (purple) are the cell's Leiden cluster (see Fig.5A-B) in the auto-encoder latent space. Ridge models (black) are penalized linear regressions combining all hand-built features or all latent dimensions, fitted on the discovery wells with a chosen penalty. The early RAMPKAR2 response to glucose is the strongest single feature (ρ = −0.17), larger in magnitude than the late glycolytic index (ρ = +0.11). Combining features raises ρ to 0.26–0.29, which corresponds to about 4% of the within-well variance in the axis (held-out Q^2).

TAKE-HOME

More glycolytic conditions are not more inflammatory. Across conditions, the glycolytic index and the inflammatory score don't line up (ρ = −0.06), contrary to the classical glycolysis–inflammation link.

Inflammatory state is decided mostly cell by cell. Stimuli set how many cells make cytokines, but about three-quarters of the variance in the inflammatory score lies between neighbouring cells in the same well.

A cell's metabolic dynamics carry a weak but real signal about its fate. **The early AMPK response to glucose is the best single predictor**, but metabolism explains only about 4% of cell-to-cell differences, so it is a modest but interesting lever for CAR-M on its own.

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